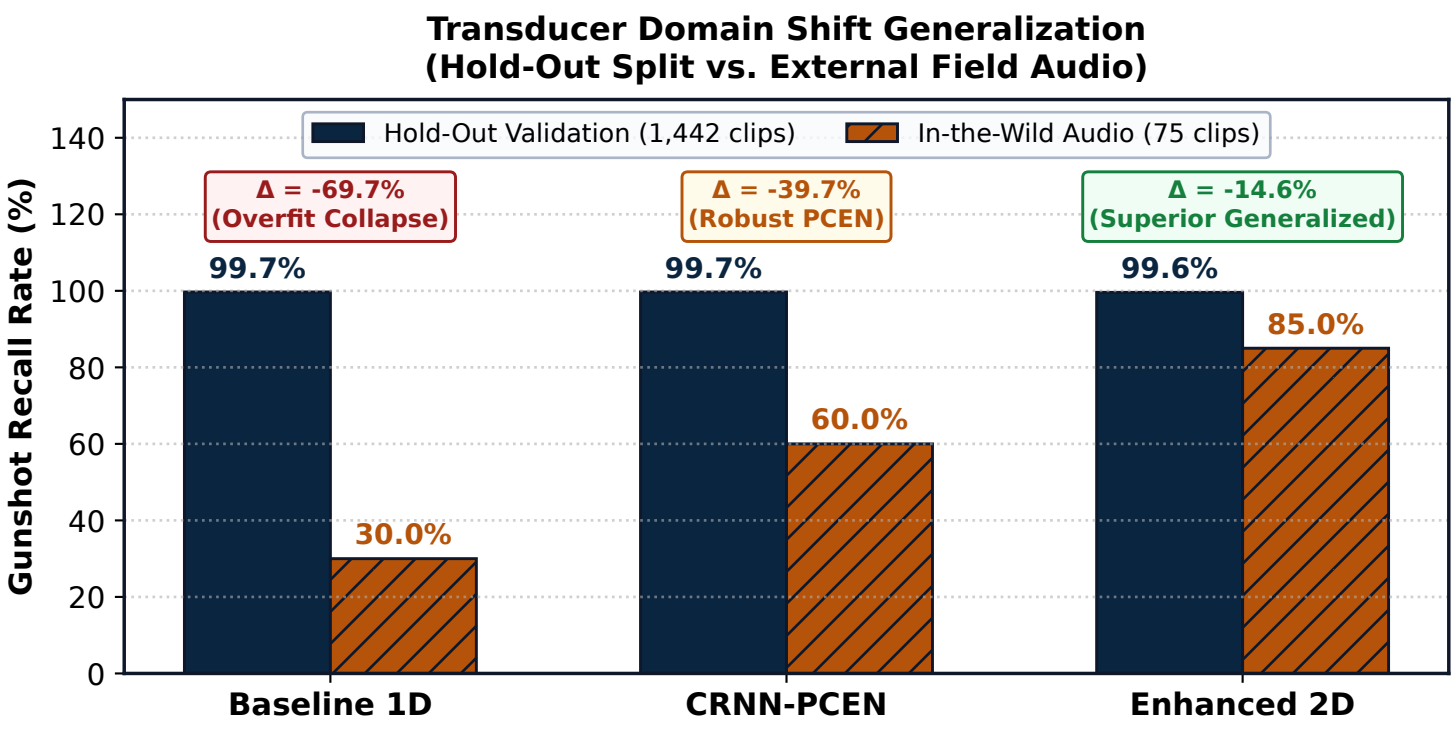
Transducer Domain Shift Generalization
(Hold-Out Split vs. External Field Audio)
Hold-Out Validation (1,442 clips)
In-the-Wild Audio (75 clips)
Gunshot Recall Rate (%)
0
20
40
60
80
100
120
140
Δ = -69.7%
(Overfit Collapse)
Δ = -39.7%
(Robust PCEN)
Δ = -14.6%
(Superior Generalized)
99.7%
30.0%
99.7%
60.0%
99.6%
85.0%
Baseline 1D
CRNN-PCEN
Enhanced 2D